

Sofia Tan

Junior Mechatronics Software Developer

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Professional Summary

Computer science graduate focused on robotics software, embedded C++, and prototype validation. Combines microcontroller work with practical web tooling for hardware teams.

Technical Skills

Arduino, ESP32, C++, JavaScript, RESTful APIs, React, Flutter, SQL

Experience Highlights

- Implemented obstacle-avoidance logic and motor control tuning for an educational robot kit using Arduino.
- Created internal dashboards in React for test-session playback, issue reporting, and experiment notes.
- Worked with technicians to debug wiring faults, noisy sensors, and intermittent power issues during prototype demos.

Selected Projects

- Smart greenhouse controller with sensor monitoring, irrigation triggers, and web-based configuration.
- Mobile app for robotics club inventory, diagnostics, and field-test reports.

Education

Bachelor-level Computer Science or Engineering background with project-based robotics, embedded, software, or automation exposure through coursework, competitions, freelance work, or internships.